



Development of wear resistant pressing moulds for the production of diamond composites

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ABSTRACT

In the powder metallurgical process, diamond material composites are produced by pressing and sintering of a mixture, consisting of diamond and metal powder like cobalt. Due to the high hardness of diamond, the surface of press die is easily damaged. To increase the life time of press dies, thin layers, deposited by PVD, has been employed. First, the standard tests, such as scratch test and pin-on-disc, were performed in order to characterize the adhesion and the tribology of the coatings. The results of standard tests were evaluated by 3D surface analyzer, SEM, and light microscope. Afterwards, PVD thin layers were deposited on the inner surface of the dies in order to analyze and compare their behavior during the compaction process. The results evidenced remarkable effect of thin PVD coatings on enhancing wear resistance of the press dies.

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1. Introduction

Sintering technology allows the production of complicated parts almost without subsequent machining and is therefore an advantageous method for production small and medium sized structural parts or components. Pressed green bodies have been sintered at a second stage in order to increase the density. Although the pressing has been optimized over the years, further improvements on wear resistance of the dies are still necessary. Solimanjad and Larsson (2003) demonstrated that the behavior of the powder during the pressing is very complex and not fully understood. It includes displacement, elastic and plastic deformation of the particles.

The friction between the powder and the die during the pressing is one of the factors which is responsible for the wear and consequently, short life time of the die. Special problems occur during pressing of diamond composites consisting of diamonds and a metallic matrix, like cobalt powder. Owing to high hardness of the diamonds in comparison to the die, the inner surface of the die is prone to damage during the compaction process. Hence, after a specific time, it is impossible to produce green bodies with the desired quality, and pressing tools must be replaced. Replacing these parts is costly and time-consuming.

A review of the existing literatures reveals that several authors tried to increase the wear resistance of the tools. For example, graded structures with cobalt content ranging from 30 to 10 wt%

from surface to the center of the mould have been prepared by Colin et al. (1993) using either solid state or liquid phase sintering. Smolik et al. (1999) have shown that the frictional wear resistance of sintered cemented carbides tools can be even increased significantly by substituting carbides by diamonds. Acchar and Wolff, 2005 suggested that alumina reinforced with refractory carbides is a good alternative to cemented carbides, for cutting tools and wear resistant applications. This sintered material shows a homogeneous distribution of WC grains in an alumina matrix, which appears to have large grains with very clean grain boundaries and thereby offer a high wear resistance. However, manufacturing of the wear resistant pressing moulds through named processes is costly and difficult to perform.

Thin PVD films have been recently employed, as an alternative, in order to increase the wear resistant of the tools. Reisel et al. (2003) have employed diamond like carbon (DLC) thin coatings as an alternative for solid lubricant in order to increase the wear resistance and life-time of tools used in warm massive forming. Lousa et al. (2001) deposited chromium/chromium nitride multilayers on die-casting tools. It was shown that these multilayer systems enhance significantly the hardness of the tool steels. In addition, using chromium inter-layer results in higher adhesion between the coating and tool steels. In our previous work (Tillmann and Vogli, 2008), it was shown that depositing multilayer PVD coating consisting of titanium and titanium aluminum nitride increases the hardness and wear resistance of the tools used in electromagnetic sheet metal forming (EMF) die. It was observed that with a decrease of the metallic layer thickness the tribological and mechanical properties of the total layer were enhanced. Since the EMF process

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Table 1
The properties of coated systems.

	Roughness		Hardness [GPa]	Thickness [μm]	Critical load [N]
	Rz [μm]	Ra [μm]			
DLC	8.18 $\pm$ 0.18	1.8 $\pm$ 0.03	5.2 $\pm$ 0.5	3.5 $\pm$ 0.2	30 $\pm$ 5.0
TiAlSN	6.33 $\pm$ 1.4	0.97 $\pm$ 0.04	27.5 $\pm$ 1.0	3.9 $\pm$ 0.2	35 $\pm$ 5.0
Ti/TiAlN	1.7 $\pm$ 0.15	0.24 $\pm$ 0.02	22 $\pm$ 1.0	3.0 $\pm$ 0.2	40 $\pm$ 4.0

differs completely from the powder compaction process, the objectives of this investigation is to adapt hard PVD multilayers and diamond like carbon (DLC) as well as nanocomposite layers for increasing life-time of the die used in manufacturing of diamond composites.

2. Experimental

In this work, three different coating systems have been deposited: monolayer, multilayer, and nanocomposites. In the category of monolayer, DLC (diamond like carbon) was deposited. Multilayer coating consists of 10 layers of titanium and titanium aluminum nitrides (Ti/TiAlN). As nanocomposite layer titanium aluminum nitrides (TiAlSN) were employed, which have a nanometer grain size and hence, have a high hardness and toughness. Dies used as substrate are of steel C60 (hardened and unhardened). E. Bemporad et al. (2006) showed that the higher thickness of coating layer results in higher thermal residual stress, and consequently, lower fracture toughness of the systems. Therefore, the thickness of the layers was limited to 4 μm .

Coating layers were deposited by CC800sinox device (CemeCon, Germany). The hardness and elasticity of coating were measured by Nano indenter Xp (MTS Nano Instruments, USA).

The scratch tests were employed in order to evaluate the adhesion of the coatings. The samples, used for this test, are made of steel C60, and had unpolished surface with a hardness of 260 HV5. They were in a rectangular form, 10 $\times$ 20 $\times$ 30 mm. Before performing the test, samples were cleaned ultrasonically with ethanol for 10 min. During the test, the force was increased linearly up to 100 N. Simultaneously, the acoustic emission was recorded. After performing the scratch test, the samples were analyzed with Alicona 3D-analyzer (Infinite Focus).

In the pin-on-disc tests, a ball, with a diameter of 5 mm, consisting of WC/Co, is pressed by 5 N load on the coated samples, which rotate with a linear velocity of 0.4 m/s. During the test, the tangential force was steadily measured. Afterwards, the friction coefficient was calculated as a fraction of the normal and tangential force, as follows:

$$\mu = \frac{F_t}{F_n} \quad (1)$$

The ambient temperature was around 25 °C with a relative humidity between 45%–60%. Before performing the test, the pin and the disc were cleaned ultrasonically for 10 min. All samples have a radius of 35 mm, and a thickness of 5 mm. In order to have a movement, as long as 1000 m on the disc, the rotational speed and track radius were exactly set up. After performing the pin-on-disc, the samples were once again ultrasonically cleaned in ethanol for 5 min. Afterwards, the surface of samples was analyzed by 3D-Analyzer. The wear track shape was analyzed at different positions in order to measure the cross-sectional area of the wear track. This measurement was performed at five different positions, and after that, the arithmetic median of the measured area was calculated. According to Archard, 1976 the wear volume is proportional to both normal force and the distance, S :

$$V = kFS \quad (2)$$

The constant factor, k , is known as specific wear rate. For pin-on-disc test results, since the track radius is larger than track width, Eq. (2) yields:

$$k = \frac{A_c}{Fn} \quad (3)$$

where A_c is the cross-sectional area of the wear track, and n is the number of revolutions. Through this equation, the specific wear rate for the different coated systems was measured.

The moulds deposited by different coating layers were filled by 2.5 g powder mixture, composed of cobalt (Eckgranules, diameter $\leq 40 \mu\text{m}$) as matrix and diamonds particles (Element Six, SDB1055 40/50 US-mesh). The powder mixture was pressed in the uni-axial press machine (Mohr and Federhoff AG) up to 50 kN. The force increased linearly from 0 to 50 kN. The morphology of the layers and the scratches on the pressing dies were characterized by using SEM (JEOL JXA-840).

3. Results and discussion

3.1. Layer characterization

The properties of three deposited systems are shown in Table 1. Since the inner surface of the die is difficult to polish, the surface was just grinded with a roughness R_a between 0.24 and 1.8. Three measurements were done on each sample and the arithmetic median was calculated. The hardness of the samples reflects their resistance against the penetration of counterparts. As it can be seen, TiAlSN layers which possess nanometer grain size are harder than Ti/TiAlN layers which contain metallic titanium inter-layers. In this work, DLC deposited on the samples has a hardness of 5.2 GPa which is lower than that of other coating layers studied in this work. The properties of amorphous carbon films can be designed through changing parameters used during deposition. Hardness of DLC film depends significantly on the ratio of sp^2 to sp^3 hybridized bonds present in the material. Graphite consists purely of sp^2 hybridized bonds, whereas diamond consists purely of sp^3 hybridized bonds. DLC deposited in this work possesses high amount of sp^2 hybridized bonds. In this study, the characterization of the adhesion of the coating layer on the substrate was performed by scratch test. Fig. 1 shows the acoustic emission recorded during the scratch test of different coating systems.

In the multilayer systems, several peaks were observed which are connected to several layers on the coating were observed. In coated samples, different kinds of failure occur. Typically, sudden increasing of acoustic emission corresponds to the adhesive or cohesive failure of the coated systems, called critical load. The critical load of DLC was the lowest and critical load of Ti/TiAlN was the highest among the test samples (Table 1). After performing the scratch test, the samples were analyzed with an optical 3D-analyzer. Fig. 2 shows the macrograph of scratch trace sections corresponding to the position of critical loads measured by acoustic emission. Obviously, spalling can be observed on the TiAlSN coating layer. No spalling was observed on DLC and Ti/TiAlN coating layer.

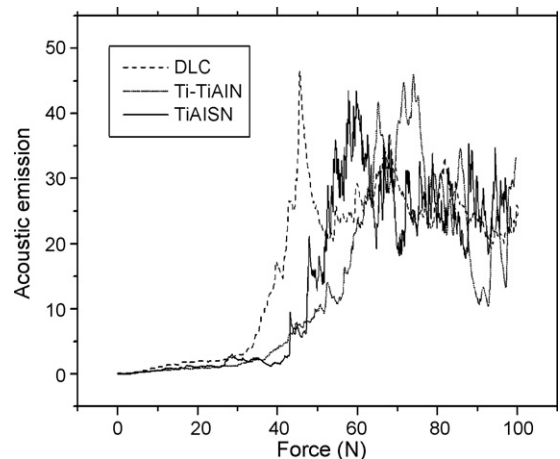


Fig. 1. Acoustic emission versus tangential force for different coating systems.

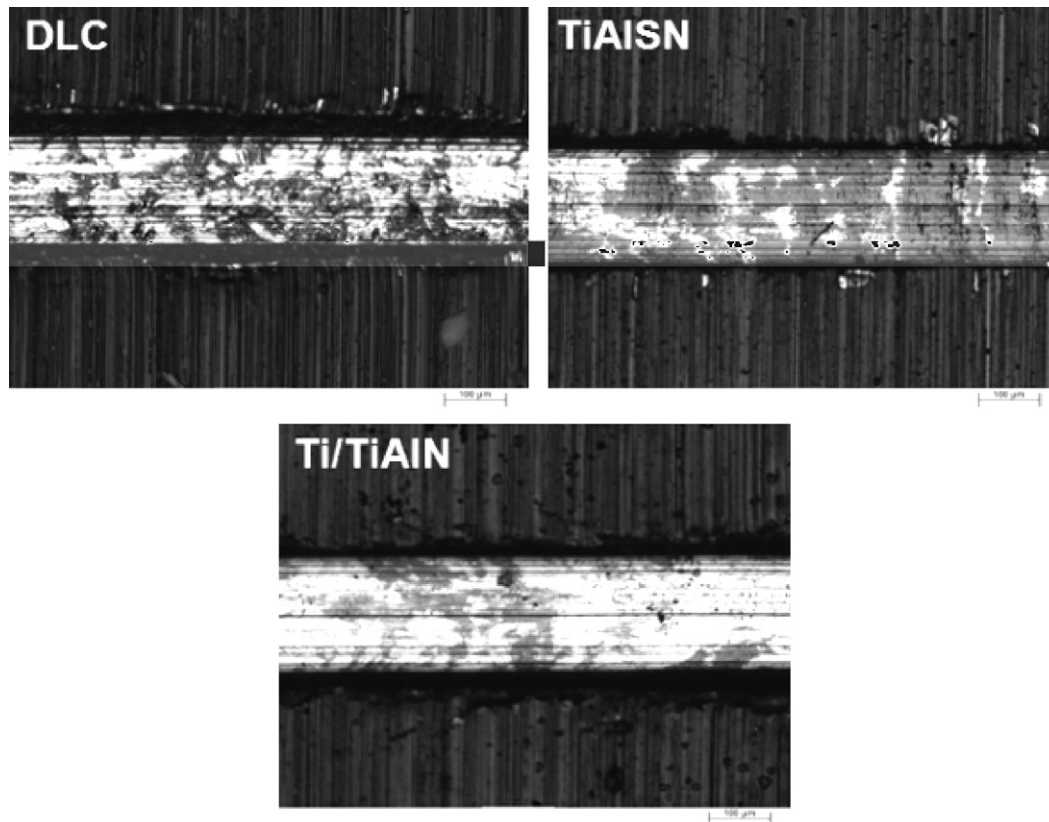


Fig. 2. Macrograph of scratch traces of different coating systems under critical load.

Fig. 3 shows the development of the friction coefficient for different coated systems during the test. Since the samples were not polished and had a high roughness similar to the inner surface of the die, the friction coefficient measured is higher in comparison to other measurements in similar study done by Su and Kao (1998). In order to show the effect of surface roughness on the friction coefficient, Ti/TiAlN was deposited on both polished and unpolished substrate.

Fig. 4 shows the overall friction coefficient of coating systems. It can be seen that DLC has the lowest friction coefficient among the samples investigated. In addition, Ti/TiAlN deposited on the polished substrate has a 14% lower friction coefficient in comparison to deposited Ti/TiAlN on the unpolished substrate. This difference is 5.3% for polished and unpolished substrate. It relies on the higher roughness of the unpolished samples compared to the substrate

polished up to 1 µm. High roughness of the disc surface results in increasing the micro contact surface and micro welding between the ball- and the disc surface during the pin-on-disc test. Therefore, the tangential force, and consequently, friction coefficient increase. In this case, running-in phase is longer, and more revolutions are needed to reach the stationary friction coefficient (Fig. 3). In a similar study, Harlin et al. (2006) have also shown that the surface roughness of the coating is critical in TiN/steel sliding contact.

The specific wear rate for the different coated systems was also measured. The obtained results are illustrated in Fig. 5. The unpolished substrate from steel C60 has the highest specific wear rate among the samples. Obviously, in addition to surface roughness and the friction coefficient, wear rate of a coated sample depends on the surface hardness. Enhancing the hardness of the surface results in decreasing the deformation of the disc under normal loading, and consequently, wear resistance of the specimen increases. For

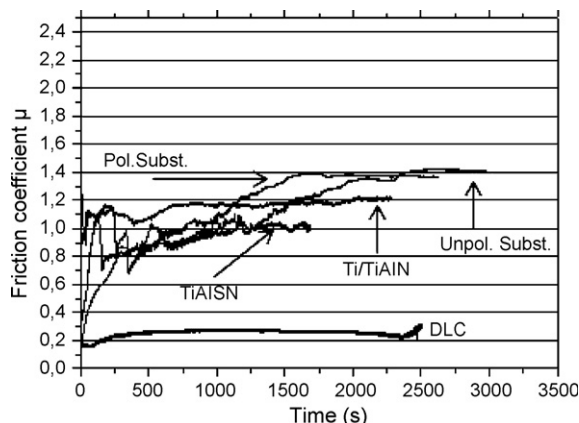


Fig. 3. Friction coefficient of the substrates and different coating systems.

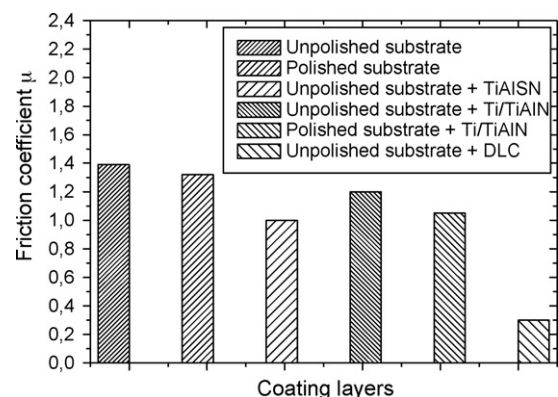


Fig. 4. Stationary friction coefficient of the substrates and different coating systems.

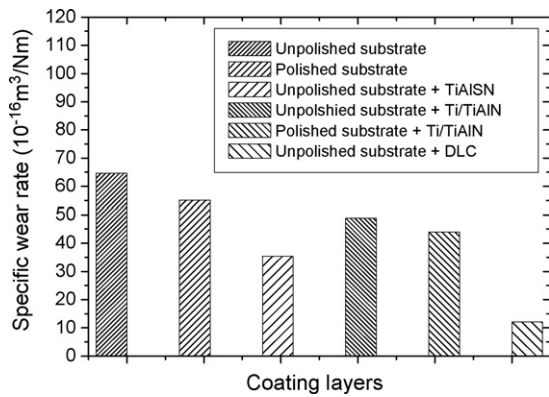


Fig. 5. Specific wear rate of the substrate and different coating systems.

example, TiAlN, which possesses higher hardness than Ti/TiAlN, has higher wear rate than that of Ti/TiAlN. However, it must be considered that DLC, despite of its lower hardness compared to Ti/TiAlN and TiAlN, possesses the lowest wear rate among all samples. Very low wear rate of the DLC is related to its extremely low friction coefficient. Namely, very low friction coefficient of DLC and the polished surface of the sample lead to decrease the tangential force between the ball and the disc. As a result, abrasion of the counter parts in the pin-on-disc test reduces. The results show that through deposition of DLC coating, the specific wear rate of the substrate decrease up to 81%.

The optical micrograph of the wear tracks are illustrated in Fig. 6. In both samples, coated by Ti/TiAlN and TiAlN, chippings can only be observed at the center of the wear track. On the sample coated with DLC, no chipping was observed. Since the friction coefficient of DLC is much lower than that of other samples, the wear rate of the ball rotating on the disc, and consequently, its worn surface is smaller than that of other specimens. Therefore, under the same loading condition, deformation and track width of the disc coated with DLC are obviously lower than wear tracks of the other coated systems. Van Essen et al. (2006) have shown that sudden increase in friction coefficient arises from extension of the chipping to the outside of the wear track. In this study, chipping was observed for TiAlN and Ti/TiAlN in the wear track, which also has no sudden increase in the friction coefficient. It means that the pin was still in contact with undamaged part of the wear track. Cho et al. (1998) have shown that no catastrophic failure occur during the first 35,000 revolution if the load employed is lower than 4–5% of the critical load measured in scratch test. This corresponds to our results. The maximum revolution in the tests described here was 14,000 for a radius of 11 mm and the pin-on-disc tests were performed with a normal force of 5 N, which is 12 times lower than critical loads of all samples. However, it should be considered that in (Cho and Lee, 2000), another counterpart (diamond indenter, radius 200 μm) with a smaller load (< 0.75 N) was used.

Due to the low friction coefficient of DLC coating, there is no chipping to be observed after 1000 m testing distance. For all scratch test samples, chippings occur in one pass, in contrast, in pin-on-disc, it takes several revolutions before any chipping occurs in pin-on-disc. This 'incubation' period by pin-on-disc test suggests that the failure of the coating layer is not catastrophic.

3.2. Coated pressing tools

During powder pressing operations, the normal- and tangential forces caused by diamonds are also much higher than during the pin-on-disc. Consequently, the chipping can be observed on the coated dies in one pass, too. The die coated by DLC has a

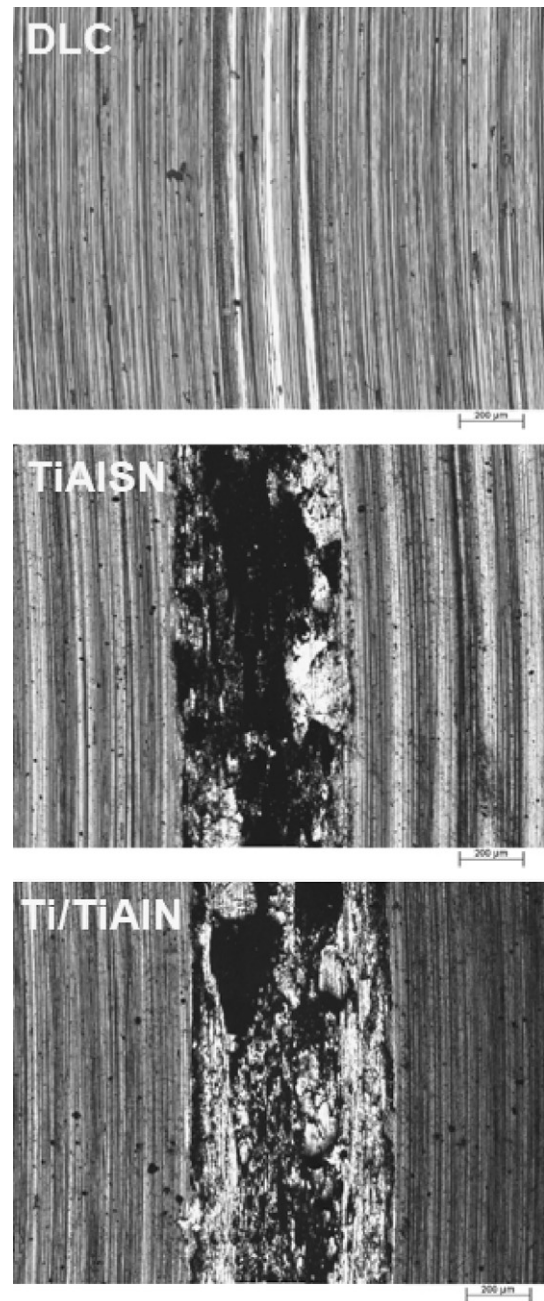


Fig. 6. Macrographs of the wear track on different coating layers in pin-on-disc test.

softer surface in comparison to the other coated systems. Therefore, the yield stress of the coating was exceeded. Consequently, the plastic deformation of the coating layer at different sites can be observed. The TiAlN coating layer possesses higher hardness than DLC. Hence, the scratches caused by diamonds on this coating layer have smaller lengths and depths in comparison to DLC. Within the dies coated by TiAlN, a large number of micro cracking was observed on the scratch trace. On the pressing tools coated by Ti/TiAlN, a large amount of cracks can be observed out of scratch trace, beginning from the both scratch sides. However, all coating systems show lower abrasion of the surface in comparison to the uncoated systems, as seen in Fig. 7. White areas on the surface of the coated moulds represent the cobalt powder remaining on the surface after pressing. Obviously, the width of scratches on the coated dies is smaller than that of uncoated dies. The thinnest scratches can

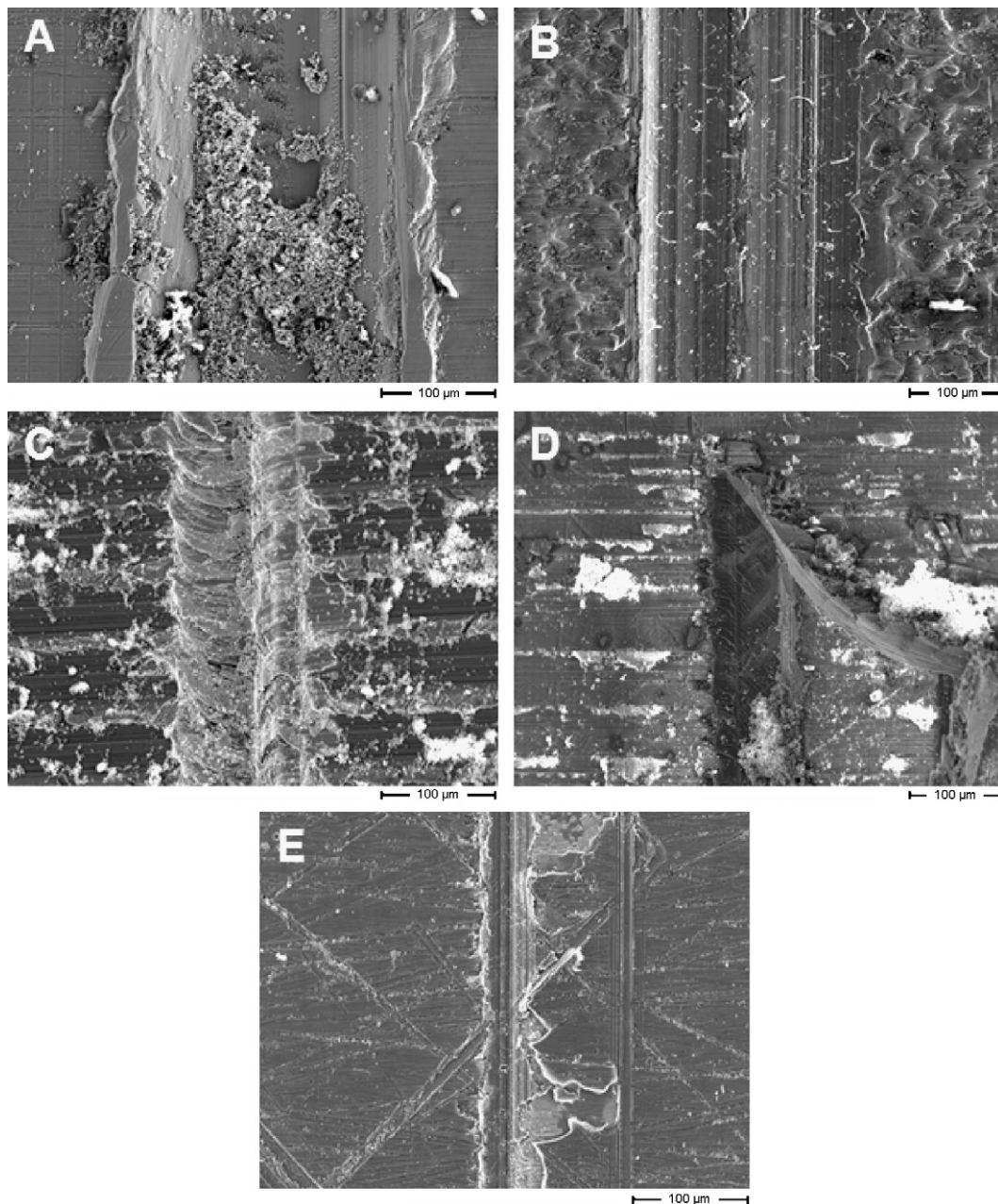


Fig. 7. Macrograph of the scratches on the surface of different pressing moulds; uncoated unhardened steel C60 (A), uncoated hardened steel C60 (B), coated by DLC (C), Coated by TiAlSiN (D), and coated by Ti/TiAlN (E).

be observed on the dies coated by Ti/TiAlN. In addition, scratches of DLC layer are generally wider than those observed on the die coated by TiAlSiN. The largest number of scratches can be observed on the dies coated by DLC layer.

4. Conclusion

In this Work, three different PVD coatings have been studied and employed on the pressing tools used.

- DLC coating layer has the lowest friction coefficient and wear rate among the systems.
- TiAlSiN has lower wear rate in comparison to Ti/TiAlN.
- Chipping is limited to the wear track center during 1000 m for all samples in pin-on-disc test.
- All samples except the one coated by DLC show chipping on the wear track during the first 1000 m revolution.
- TiAlSiN and Ti/TiAlN show respectively 16.6% and 33.3% higher critical load in comparison to DLC.
- In scratch tests, spalling was only observed in samples coated by TiAlSiN outside the wear trace. This kind of failure did not occur by the other samples.
- DLC layer deposited in this work had more ductile behavior compared to other coating systems.
- TiAlSiN and Ti/TiAlN are harder than DLC. Therefore, they show a lower surface deformation and a higher amount of cracking after pressing diamond composites.
- The scratches observed on the dies coated by Ti/TiAlN layer are thinner than those on the dies coated by DLC and TiAlSiN layers.
- Both TiAlSiN and Ti/TiAlN have significantly increased the life time of the pressing dies.

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